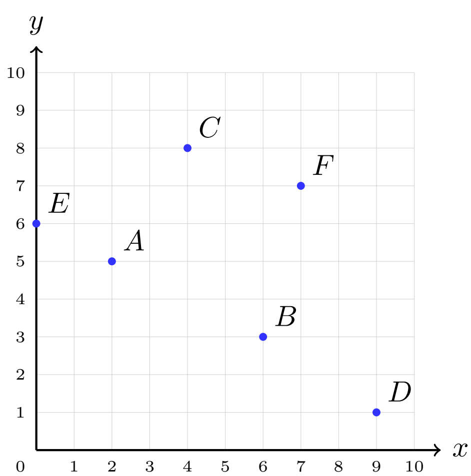
Reading and Plotting Points on a Grid

Reading ordered pairs off a first-quadrant grid, plotting new points on it, and turning two number patterns into points

How to use this sheet.

- An *ordered pair* is written (x, y). The first number tells you how far to travel **across** from the corner marked 0, and the second tells you how far to travel **up**. Across first, then up — every time, in that order.
- Every grid on this sheet starts at 0 in the bottom-left corner and counts by ones in both directions.
- Say the journey out loud as you go: "start at zero, across four, up eight." The one mistake this sheet keeps coming back to is doing those two steps in the wrong order.
- Use a ruler or the edge of a card to trace across and up. It is much easier than counting squares by eye.

1. **Read point A.** Start at the corner. Trace across to A, then up to A.



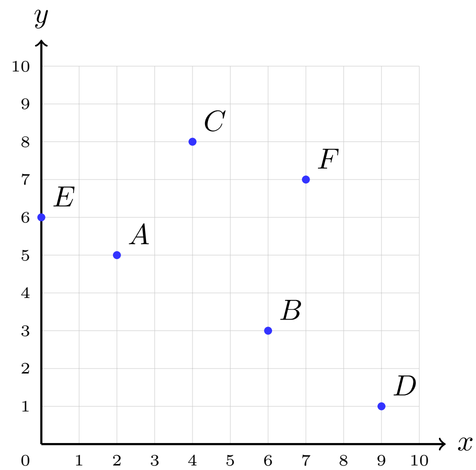
Point A is marked in blue with five other points.

How far across is A from 0? ________

How far up is A from 0? ________

Write it as an ordered pair: A = (________, ________)

2. **Read point B.** Same grid, same method.



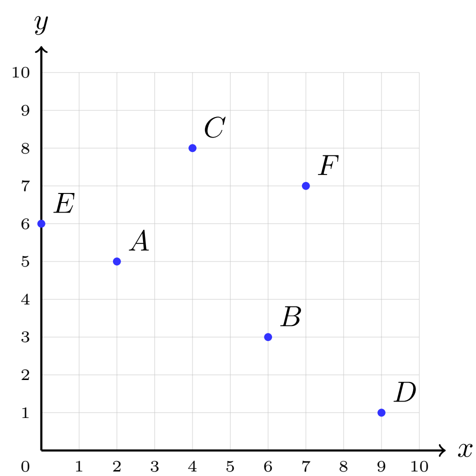
Across __________ Up __________

$B = ($ __________, __________$)$

Now write C and D as ordered pairs too, for practice:

$C = ($ ____ , ____ $)$ $D = ($ ____ , ____ $)$

3. **A point that sits on a line.** Point E is not out in the middle of the grid — it is sitting right on one of the two arrows.



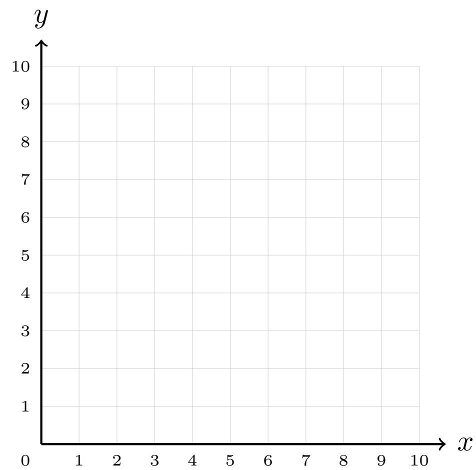
How far across is E from the corner? __________

How far up is E? __________

$E = ($ __________, __________$)$

Which arrow is E sitting on? ________________

4. Plot two points, then measure across. Plot $P(3,6)$ and $Q(8,6)$ on the empty grid, label them, and join them with a straight line.



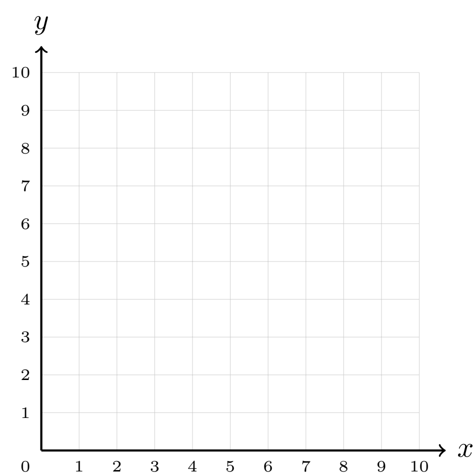
Both points are the same height up. So the line you drew is flat.

x of Q ________ $-$ x of P ________

How many squares long is $\overline{PQ}$? ________

Why did you not need the y-coordinates to answer that? ____________________

5. Now measure up. Plot $R(4,2)$ and $S(4,9)$, label them, and join them.



This time both points are the same distance across, so the line is upright.

y of S ________ $-$ y of R ________

How many squares long is $\overline{RS}$? ________

Which coordinate is the same for both points?

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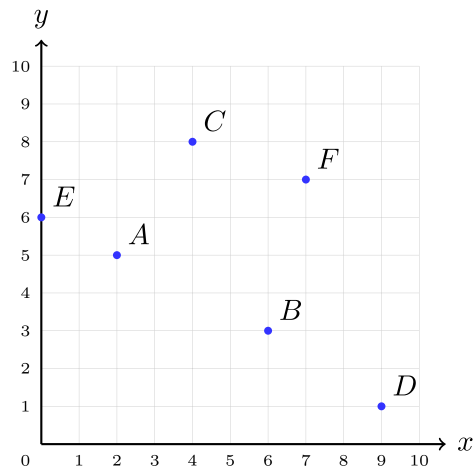
6. How much higher? Look again at the first grid. Point C is at $(4, 8)$ and point B is at $(6, 3)$.



"How much higher" is a question about going *up*, so it is the second number in each pair you need.

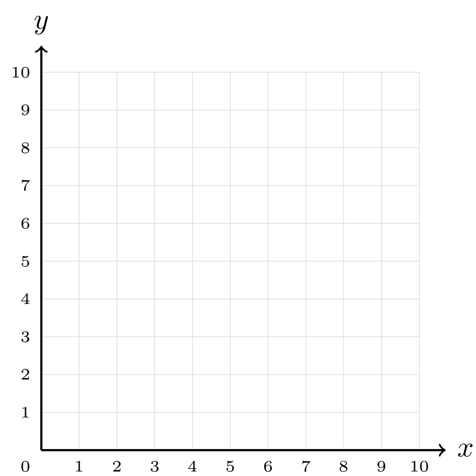
y of C _________ − y of B _________

C is _________squares higher than B.

Careful: B is further across than C. Does that change how much higher C is? ____________

7. Two rules make points. Rule A: *start at* 0 *and add* 1 *each time.* Rule B: *start at* 0 *and add* 2 *each time.* Take the first number from Rule A and the second from Rule B, and you get an ordered pair.

Rule A (x)	0	1	2	3	4
Rule B (y)	0	2	_________	_________	_________



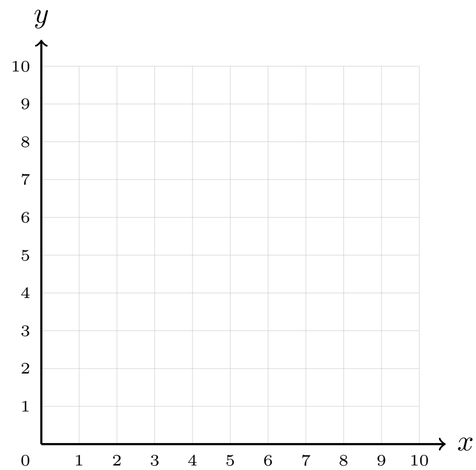
Plot all five ordered pairs on the grid and join them in order.

y when $x = 3$: _________

y when $x = 4$: _________

Finish the sentence: every y is always _______ times the x beside it.

8. Keep the pattern going. Use the same two rules as problem 7. What is the next ordered pair after $(4, 8)$?



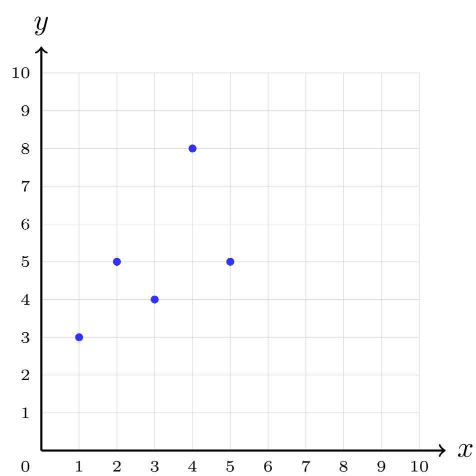
Rule A adds 1, so the next x is ________

Rule B adds 2, so the next y is ________

The next point is (________, ________). Plot it on the grid.

Does the new point sit on the same straight line as the others? ____________

9. A graph that tells a story. Each point shows how many books Maya read in one week. Week 1 is at $(1, 3)$, and the rest follow.



Read the height of each point and write it down.

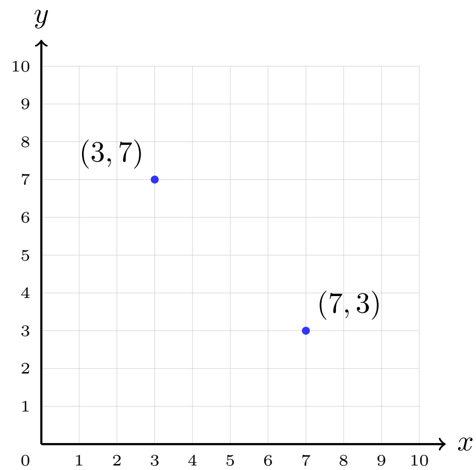
Wk 1 ________ Wk 2 ________ Wk 3 ________ Wk 4 ________ Wk 5 ________

Books altogether: ________

Share them equally over the five weeks: mean ________books per week

Which week was Maya's best? __________

10. Explain it to a friend. Nia says: "$(3, 7)$ and $(7, 3)$ must be the same point, because they use the same two numbers." Both points are marked on the grid.



Trace the journey to each one with your finger, saying it out loud.

Then write two or three sentences telling Nia why the two points are not the same, and what the *order* of the two numbers is for.

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